COCKROACHES

Cockroaches have been around since the time of dinosaurs!



American Cockroach
The American Cockroach is the largest cockroach found in houses. Females can hatch up to 150 offspring per year. Cockroaches don't develop wings until they become adults.

German Cockroach
German cockroaches can be found all over the world. They are the most common cockroach in the United States. Each German cockroach can live about 100-200 days.

Brown-banded Cockroach
The wings of male cockroaches are larger than the female's wings. Brown-banded cockroaches often hide their eggs in or under furniture. They usually live for 5-6½ months.

Oriental Cockroach
They are actually from Africa. They are large and very dark compared to other cockroaches. They usually travel through sewer pipes and drains. They prefer dirty places and cooler temperatures than other cockroaches.

Viviparous Cockroach
Diploptera punctata is a species of cockroach in the family Blaberidae. It is one of the few cockroach species that is viviparous. Adults are chemically defended, having a modified tracheal gland and spiracle on each side which squirts quinones which can poison or discourage a predator.

Facts

- *Cockroaches are known to carry diseases like dysentery, typhoid and poliomyelitis, as well as gastroenteritis.*
- *According to The National Cooperative Inner-City Asthma Study (NCICAS) – 23 percent to 60 percent of urban residents with asthma are sensitive to the cockroach allergens.*
- *Cockroaches have been implicated in the spread of 33 kinds of bacteria, including E. coli and Salmonella species, six parasitic worms and more than seven other types of human pathogens.*



Diploptera punctata, a viviparous cockroach, produces a nutritionally dense crystalline "milk" to feed their live-born young. It is found in Myanmar, China, Fiji, Hawaii, and India. Scientists think Cockroach milk could be the super food of the future.

4.3 Frog - Rana hexadactyla

About 360 million years ago, amphibians were the first vertebrates to live on land. Amphibians are diverse, widespread, and abundant group since the early diversification. There are about 4,500 species of amphibians. Frog is an amphibian and hence placed in the class Amphibia [Greek. *Amphi* - Both, *bios* – life]. The largest order, with more than 3,900 species, is Anura, which includes the **frogs** and **toads**. *Rana hexadactyla* is placed in the order Anura. Frogs live in fresh water ponds, streams and in moist places. They feed on small animals like insects, worms, small fishes, slugs, snails, etc. During its early development a frog is fully aquatic and breathes like a fish with gills. It is **poikilothermic**, i.e., their body temperature varies with the varying environmental temperature.

Morphology of Frog

Classification

Phylum	:	Chordata
Class	:	Amphibia
Order	:	Anura
Genus	:	*Rana*
Species	:	*hexadactyla*

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